NATIONAL MATHEMATICS CONTEST
PRIMARY-LEVEL PAPER 2 SOLUTIONS.

SECTION A : 5 marks each

1. The sum of the digits of 2023 is $2 + 0 + 2 + 3 = 7$ which is a prime number. What is the next year that has the sum of its digits equal to a prime number?

Solution:

It is obvious that only the last digit(s) should be changed. The prime number following 7 is 11 (remember 9 is not prime), so the next year that has the sum of its digits equal to a prime number is in 4 years, it is 2027.

2. In this problem, each digit is used once and only once. From the digits $5, 6, 7, 8, 9$ a two digit number is subtracted from a three digit number. What is the smallest difference?

Solution:

The smallest difference occurs when the largest two digit number is subtracted from the smallest three digit number. The result is $567 - 98 = 469$.

3. Scovia challenged Sarah to a 200m race. For every 5m that Scovia runs, Sarah can only run 4m. When Scovia crosses the finish line, how many metres will Sarah have run?

Solution:

When Scovia runs 200m, this is equivalent to running 5m a total of 40 times. While She does this, Sarah will run 4m a total of 40 times. Therefore Sarah will run $4 \times 40 = 160$m in all when Scovia crosses the finish line.

4. In the sequence below, what will be the 2024th symbol?

$$\rightarrow, \downarrow, \leftarrow, \uparrow, \rightarrow, \downarrow, \leftarrow, \uparrow, \rightarrow, \downarrow, \leftarrow, \ldots.$$

Solution:

The 4th symbol is $\uparrow$, and the sequence cycles after every 4 symbols. Then the 8th,12th,16th, and so on, up to the 2024th symbol are all $\uparrow$. Then the 2024th symbol is $\uparrow$.

5. There were 10 runners in the finals of a race. Immaculate beat 5 persons more than the number of persons that defeated her. In which place did Immaculate finish?

Solution:

Other than Immaculate, there were 9 runners in the race. Since Immaculate beat 5 persons more than those that beat her, she finished ahead of 7 runners and behind 2. Immaculate finished in the 3rd place.

6. There are 60 learners in a class. Always two students share a desk. Every boy shares a desk with a girl. Exactly half the girls share a desk with a boy. How many boys are in the class?

Solution:

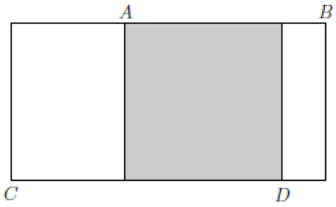
There are as many boys as half the girls (because they share desks). Boys are therefore a third of the class which is $\frac{1}{3} \times 60 = 20$.

7. Martha is 2 years less than half of Benjamin's age. In 4 year's time Benjamin would be 3 times Martha's age 2 years ago. What is Benjamin's age?

Solution:

Let Benjamin's age be x years. Then Martha is $(\frac{x}{2} - 2)$ years. In 4 years time, Benjamin will be $x + 4$ years . However, 2 years ago was $\frac{x}{2} - 4$ years. Thus $x + 4 = 3\left(\frac{x}{2} - 4\right) \implies 2x + 8 = 3x - 24 \implies x = 32$. The current age of Benjamin is 32 years.

8. In the figure on the right, there is a shaded square inside a rectangle as shown. From A to B is 6 cm and from C to D is 8 cm. What is the perimeter of the larger rectangle?



Solution:

Let the length of one side of the square be x. Then $x + a = 6$ and $x + b = 8$ where the length of the rectangle is $b + x + a$ and the width is x. The half-perimeter is $b + x + a + x = b + 2x + a = (x + a) + (x + b) = 6 + 8 = 14$. The perimeter is therefore 14 cm +14 cm = 28 cm.

9. At Blessed primary school, there are 100 pupils who are participating in 2024 National Mathematics Contest organised by UMS of which 99% are girls. Some of the pupils live at the school (boarders) while the rest are day scholars. Of those boarding, 98% are girls. How many girls are day scholars?

Solution:

It is clear that the number of girls is 99 and the number of boys is 1. This boy lives at school otherwise all the pupils living at school(boarding)would be girls. The 2% boys living at school is accounted for by this 1 boy. Therefore the 100% pupils living at school is accounted for by 50 pupils (49 girls and 1 boy). Therefore a total of 50 girls are day scholars.

10. On a certain planet, the number of days in a week is the same as the number of weeks in a month and the number of months in a year is twice the number of days in a month. On this planet, there are 800 days in a year. How many months are there in a year?

Solution:

Let d be the number of days in a week and m be the number of months in a year. From the information we have, $d \times d = d^2$ is the number of days in one month. The number of days in a year is thus $md^2 = 800$. Also since the number of months in a year is twice the number of days in a month, it implies that $m = 2d^2 \implies d^2 = \dfrac{m}{2}$.

It then follows that $md^2 = m \times m/2 = 800$. This gives $m^2 = 1600$ hence $m = 40$. On this planet, a year has 40 months.

SECTION B : 10 marks each

11. In a certain school, the pupils in primary Five are holding an election for the class monitor. Four candidates are competing for the position. Whoever gets the highest number of votes wins. All the four candidates received a combined total of 83 votes. What is the smallest number of votes the winner could receive?

Solution:

We observe that $\frac{83}{4} = 20.75$ and so the winner must get 21 or more votes. If person 1 receives 21 votes, 62 votes must be shared between 3 persons and since $20 + 20 + 20 = 60$, at least one other person would get 21 or more votes and so the winner must get more than 21 votes. If person 1 gets 22 votes, the other three persons could get $20, 20, 21$ (for example) and person 1 wins. Therefore the smallest number of votes for the winner is 22.

12. (a) In a cinema hall, Pamella is seated in the 7th row from the front and the 7th column from the right. She is also seated in the 16th row from the back and the 14th column from the left. How many seats were occupied if the last 8 rows were not occupied in the cinema?

Solution:

The total number of columns in the cinema hall is $14+7-1 = 20$. The total number of rows in the cinema hall is $16+7-1 = 22$. A total of $(20 \times 22) = 440$ people are required to fill up the entire Cinema hall when each one of them takes a seat. If the last 8 rows were not occupied in the cinema hall, the number of occupied seats are $440 - (8 \times 20) = 440 - 160 = 280$.

(b) Before a certain regional tournament, She Cranes had won 45% of their netball games. During that regional tournament, they won 6 games and lost 2. At that point, they had won half of their games. How many games had the She Cranes team played altogether after the tournament?

Solution:

Let x be the total number of games before the regional tournament. The She Cranes have $0.45x$ wins, therefore the rest $0.55x$ are losses. But after the regional tournament, they won 6 and lost 2 more games. We can solve for x by forming the equation $0.45x + 6 = 0.55x + 2$. Solving the equation gives us $0.10x = 4 \implies x = 40$. Thus the total games (which includes the games after the regional tournament) is $8 + x = 8 + 40 = 48$.

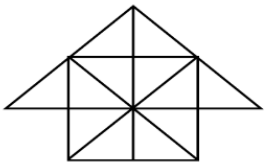
Alternative solution:
Let x be the total games played before the regional tournament. This means, before the regional tournament, She Cranes had $0.45x$ wins and $0.55x$ losses. After the tournament, She cranes had $0.45x + 6$ wins and $0.55x + 2$ losses with $x + 8$ games as the total after the regional tournament. With half of the games played being wins after the regional tournament, this results in an equation $0.45x + 8 = \frac{1}{2}(x + 8)$. After simplifying, we obtain $0.45x + 6 = 0.5x + 4$ resulting in $x = 40$. So She Cranes had played $40 + 8 = 48$ games after the regional tournament.

13. In a market A, 4 red apples are sold for 1 USD while 3 green apples are sold for 1 USD. In a market B, 4 red apples are sold for 1 USD while 6 green apples are sold for 1 USD. Daphine spends 10 USD buying m red apples and n green apples from market A and m red apples and n green apples from market B. How many apples (altogether) did Daphine buy?

Solution:

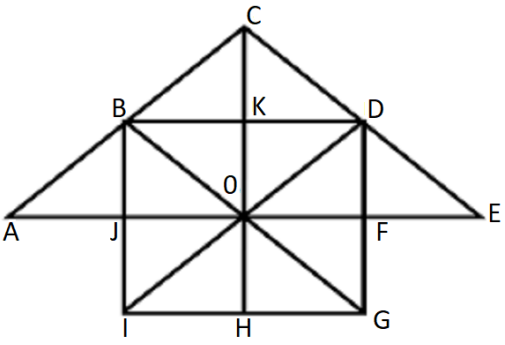
The costs for a red apple from market A is $\frac{1}{4}$USD. The costs for a green apple from market A is $\frac{1}{3}$USD. The cost for a red apple from market B is $\frac{1}{6}$USD. The cost for a green apple from market B is $\frac{1}{6}$USD. Buying m red apples and n green apples from markets A and B. Daphine spends, in dollars, $\frac{m}{4} + \frac{n}{3} + \frac{m}{4} + \frac{n}{6} = 10$.
Multiplying through by 12 gives $3m + 4n + 3m + 2n = 120 \implies 6(m + n) = 120$. Solving this equation gives $m + n = 20$. The total number of apples bought is $2(m + n) = 40$.

14. How many triangles are there in the figure given on the right?



Solution:

Let us introduce some labels in the figure as shown below.



The simplest triangles are $JBO, BKO, KDO, DFO, FGO, GHO, HIO, IJO, ABJ, BCK, CKD$ and DEF i.e., 12 in number.

The triangles composed of two components each are $IBO, BDO, DGO, GIO, ABO, CDO, CBO, CBD$ and DEO i.e., 9 in number.

The triangles composed of four components each are IBD, BDG, DGI, GIB, ACO and COE i.e., 6 in number.

There is only one triangle i.e., ACE composed of eight components.

Hence, there are $12 + 9 + 6 + 1 = 28$ triangles in the given figure.

15. In one month three Sundays were on even dates. What day of the week was the 20th day of the same month?

Solution:

Because three Sundays are on even days in this month, there are 5 Sundays in this month and the order is E, O, E, O, E where E represents even and O represents odd. If the first Sunday is on the 2nd then the last Sunday is on the 30th. If the first Sunday is on the 4nd then the last Sunday is on the 32nd. Since no month has 32 days, the first Sunday must be on the 2nd. The 6th of the month will therefore be a Thursday and so is the 20th.

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